Bo Shang, Ph.D.

Postdoctoral Associate, University of Maryland

Civil and Environmental Engineering / Maryland Transportation Institute

New York, NY

bshang@umd.edu | +1 (929) 520-0488 | Website | Google Scholar | ORCID

Robotics and AI researcher focused on reliable multimodal perception and field robotics for transportation and infrastructure. Research connects sensor integration and field data collection with computer vision, embedded control, cross-site model evaluation, and human review of engineering evidence.

Research Interests

Physical AI and field robotics; LiDAR-camera perception; vulnerable-road-user sensing and transportation safety; robotic infrastructure inspection; UAVs, visual servoing, and GPS-denied navigation; reproducible evaluation across sensors, sites, and operating conditions.

Academic and Research Appointments

University of Maryland | Jul 2026 - Present

Postdoctoral Associate, Civil and Environmental Engineering / Maryland Transportation Institute | College Park, MD

- Support hardware and software development, integration, and testing for two sponsored transportation research projects under Dr. Terry Yang.
- Conduct system troubleshooting, support field data collection, and prepare technical documentation and project deliverables.

The City College of New York, CUNY | Jul 2025 - Jun 2026

Postdoctoral Scholar, AI & Mobility Research Lab | New York, NY

- Developed roadside LiDAR and camera workflows for traffic monitoring, multimodal perception, and transportation safety research.

The City College of New York, CUNY | Jan 2025 - Jun 2025

Graduate Research Assistant, Civil Engineering | New York, NY

The City College of New York, CUNY | Dec 2022 - Dec 2024

Postdoctoral Researcher, Robotics Lab | New York, NY

- Developed robotic inspection software, structural-defect analysis, and engineering review workflows connecting sensors, robots, and cloud services.

Vaughn College of Aeronautics and Technology | Jun 2024 - Aug 2024

Instructor, Computer Engineering Summer Academy (AI module) | New York, NY

Vaughn College of Aeronautics and Technology | Jan 2024 - May 2024

Adjunct Faculty | New York, NY

The City College of New York, CUNY | Aug 2023 - Dec 2023

Adjunct Assistant Professor | New York, NY

Missouri University of Science and Technology | Jan 2020 - Nov 2022

Postdoctoral Fellow | Rolla, MO

- Developed UAV guidance and robotic deployment components for bridge inspection and contributed to multimodal infrastructure sensing research.

University of California, Merced | Jan 2016 - Aug 2017

Lecturer | Merced, CA

- Supported teaching in mechatronics, engineering computing, and unmanned aerial systems; developed initial laboratories for the UAS course.

University profile

University of California, Merced | Aug 2015 - Sep 2017

Junior Specialist | Merced, CA

Education

Northeastern University, China | 2013 - 2020

Ph.D., Pattern Recognition and Intelligent Systems

University of California, Merced, USA | 2015 - 2017

Exchange doctoral research

Northeastern University, China | 2011 - 2013

M.Eng., Pattern Recognition and Intelligent Systems

Northeastern University, China | 2007 - 2011

B.Eng., Automation

Research and Engineering Projects

Roadside LiDAR and camera perception, CUNY | Nov 2024 - Jun 2026

- Developed a workflow spanning sensor capture, temporal synchronization, spatial calibration, background subtraction, object segmentation, and road-user detection.
- Built multi-frame vehicle reconstruction, labeling and quality-assurance workflows, and model evaluation across sites with different beam counts, frame rates, and mounting geometries.
- Separated site-specific configuration from reusable processing and investigated cross-configuration generalization for traffic monitoring and vulnerable-road-user safety.

Robotic inspection automation and structural-defect analytics, CCNY | Dec 2022 - Dec 2024

- Developed CNN-based analysis of cracks, spalling, and stains, with AWS inference and a WebODM-based workflow for 3D reconstruction, localization, visualization, and measurement.
- Built an iOS application for controlling a climbing inspection robot and integrated human review into inspection-data workflows.

Bridge Inspection Robot Deployment System (BIRDS), Missouri S&T | 2020 - 2022

- Developed UAV and clamping-robot components for bridge inspection, including vision-based girder detection, embedded perception, and control of the deployment mechanism.
- Integrated NVIDIA Jetson, Raspberry Pi, ROS, embedded Linux, cameras, and robot communication; tested autonomous guidance, clamping, and traversal workflows.

UAV visible, infrared, and hyperspectral sensing | 2020 - 2022

Developed Python and Ubuntu workflows for multimodal infrastructure inspection, sensing, and flight-path planning.

Robot-assisted underwater acoustic imaging for bridge scour evaluation | 2020 - 2021

Contributed Python/C++, ROS, Arduino, embedded control, and computer-vision components to underwater inspection research.

Vision-based UAV servoing and fractional-order control | 2015 - 2019

Developed and evaluated visual guidance and control methods using cameras, embedded Linux, Raspberry Pi, and fractional-order controllers.

SmartCaveDrone and GPS-denied exploration | 2015 - 2017

Contributed RGB-D mapping, visual SLAM, and sense-and-avoid capabilities for UAV exploration and robotic archaeology.

Early robotics projects | 2010 - 2014

- 2013-2014: AR.Drone and ROS quadrotor object tracking.
- 2012-2013: Indoor LiDAR-equipped quadrotor; 2012: camera-equipped quadrotor.
- 2011: Camera-equipped fixed-wing UAV; 2010: infrared line-following smart car.

Independent Research Software and Educational Platforms

ScholarMind AI / Merit Point Academy | 2026 - Present

Independent AI product development

- Developed a bilingual AI education and research platform with a research assistant, course delivery, robotics evidence pages, and knowledge-graph learning tools.
- Directed AI-assisted software development through requirements, implementation, testing, privacy and attribution checks, and deployment.

Reproducible AI evaluation | 2026

Built evaluation workflows using versioned configurations, replay, failure analysis, and regression checks for research software and tool-using AI agents.

Teaching and Mentoring

University of California, Merced | 2016 - 2017

Teaching support for ME 190 Mechatronics, ME 143 Unmanned Aircraft Systems, and ME 021 Engineering Computing (Fortran and MATLAB); initial laboratory development for the UAS course.

The City College of New York, CUNY | 2023

ENGR 204 Electric Circuits, as Adjunct Assistant Professor.

Vaughn College of Aeronautics and Technology | 2024

MCE 355 Robotics, Mechanics and Control; SBC 012 Principles of AI; SBC 014A Principles of Research-AI. Developed and taught a five-lecture, five-lab AI module for the Computer Engineering Summer Academy.

Research mentoring and robotics outreach

Tutored high-school research assistants at CCNY (2023-2024), coached FIRST Robotics Competition participants (2022), and judged VEX Robotics Competition events for middle-school, high-school, and college students (2024).

Journal Articles

[J1] Bo Shang, Yiqiao Li. **Drive-by Urban Freight Sensing: Large-Scale Fine-Grained Truck Classification Using Google Street View Imagery and Deep Learning.** *IEEE Open Journal of Intelligent Transportation Systems, vol. 7, pp. 2327-2347*, 2026. Published September 4, 2026; open access. DOI PDF Record

[J2] Jinglun Feng, Bo Shang, Ejup Hoxha, César Hernández, Yang He, Weihan Wang, Jizhong Xiao. **Robotic Inspection and Data Analytics to Localize and Visualize the Structural Defects of Concrete Infrastructure.** *IEEE Transactions on Automation Science and Engineering, vol. 22, pp. 22324-22336*, 2025. Journal paper selected for presentation at IEEE IROS 2025. DOI Record

[J3] Haibin Zhang, Zhenhua Shi, Liujun Li, Pu Jiao, Bo Shang, Genda Chen. **Code-specified early delamination detection and quantification in a RC bridge deck: passive vs. active infrared thermography.** *Journal of Civil Structural Health Monitoring, vol. 15(1), pp. 227-244*, 2025. Issue year 2025; first published online August 6, 2024. DOI

[J4] Ejup Hoxha, Jinglun Feng, Agnimitra Sengupta, David Kirakosian, Yang He, Bo Shang, Ardian Gjinofci, Jizhong Xiao. **Contrastive learning for robust defect mapping in concrete slabs using impact echo.** *Construction and Building Materials, vol. 461, article 139829*, 2025. DOI SSRN

[J5] Genda Chen, Liujun Li, Haibin Zhang, Zhenhua Shi, Bo Shang. **Aerial Nondestructive Testing and Evaluation (aNDT&E).** *Materials Evaluation, vol. 81(1), pp. 67-73*, 2023. DOI PDF

[J6] B Shang, J Liu, Y Zhang, C Wu, YQ Chen. **Fractional-order flight control of quadrotor UAS on vision-based precision hovering with larger sampling period.** *Nonlinear Dynamics*, 2019.

[J7] B Shang, C Wu, Y Hu, J Yang. **An algorithm of visual reconnaissance path planning for UAVs in complex spaces.** *Journal of Computational Information Systems*, 2014.

[J8] C Gao, C Wu, Y Zhang, B Shang, T Meng. **Research on remote image/data transmission based on TCP/IP protocol.** *Journal of Mechanical & Electrical Engineering, vol. 28(9)*, 2011.

Conference and Workshop Papers

[C1] Bo Shang, Yiqiao Li. **Roadside LiDAR for Cooperative Safety Auditing at Urban Intersections: Toward Auditable V2X Infrastructure Intelligence.** *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, pp. 757-765*, 2026. DriveX Workshop on Foundation Models for Autonomous Driving; archival proceedings. arXiv PDF Record

[C2] Bo Shang, Yiqiao Li, Arian Golrokh Amin, Camille Kamga, Jie Wei. **AI-Enhanced Sensing for Vulnerable Road User Safety at Signalized Intersections: A Survey.** *Procedia Computer Science, vol. 265, pp. 350-357 (MobiSPC 2025)*, 2025. DOI

[C3] Bo Shang, Yiqiao Li, Jie Wei, Camille Kamga. **Investigating the LiDAR Beam Count Needed for Effective Vulnerable Road User Detection.** *Procedia Computer Science, vol. 265, pp. 358-365 (MobiSPC 2025)*, 2025. DOI

[C4] Johnny Li, Bo Shang, Iresh Jayawardana, Genda Chen. **Hardware-in-the-loop and Digital Twin Enabled Autonomous Robotics-assisted Environment Inspection.** *2023 6th International Symposium on Autonomous Systems (ISAS), pp. 1-5*, 2023. DOI PDF

[C5] Liujun Li, Genda Chen, Bo Shang. **Mixed Reality Enabled Digital Twin for Robot-Assisted Bridge Element Inspection and Maintenance.** *8th World Conference on Structural Control and Monitoring (8WCSCM)*, 2022.

[C6] Z Shi, B Shang, H Zhang, L Li, G Chen. **Evaluation of User-Friendliness of Several UASs for Bridge Inspection.** *8th World Conference on Structural Control and Monitoring (8WCSCM)*, 2022.

[C7] H Zhang, P Jiao, L Li, Z Shi, B Shang, G Chen. **Delamination Detection of Concrete Bridge Slab Through UAV-Based Thermal Scanning.** *8th World Conference on Structural Control and Monitoring (8WCSCM)*, 2022.

[C8] Pu Jiao, Bo Shang, Liujun Li, Genda Chen. **The ceiling effect and flight insight of unmanned aerial vehicles during proximity inspection of bridges via computational fluid dynamics modeling and simulations.** *Structural Health Monitoring 2021: Proceedings of the 13th International Workshop on Structural Health Monitoring, pp. 866-874*, 2021. Proceedings year 2021; workshop held March 15-17, 2022. DOI Record

[C9] B Shang, C Wu, YQ Chen. **Neighborhood Optimization Method for Shaping Bode Plot With Larger Phase Margin.** *ASME IDETC/CIE*, 2019.

[C10] B Shang, C Wu, Y Zhang, YQ Chen. **Fractional order flight control of quadrotor UAS: a Simscape benchmark environment and a case study.** *IEEE CSAA Guidance, Navigation and Control Conference (CGNCC)*, 2018.

[C11] B Shang, Y Zhang, C Wu, YQ Chen. **Fractional order flight control of quadrotor UAS: an OS4 benchmark environment and a case study.** *15th International Conference on Control, Automation, Robotics and Vision (ICARCV)*, 2018.

[C12] G Zhang, B Shang, YQ Chen, H Moyes. **SmartCaveDrone: 3D cave mapping using UAVs as robotic co-archaeologists.** *International Conference on Unmanned Aircraft Systems (ICUAS)*, 2017.

[C13] B Shang, C Wu, Y Zhang, YQ Chen. **Analysis of maximum possible sampling period for a real-time vision-based control system.** *ASME IDETC/CIE*, 2017.

[C14] B Shang, J Liu, T Zhao, YQ Chen. **Fractional order robust visual servoing control of a quadrotor UAV with larger sampling period.** *International Conference on Unmanned Aircraft Systems (ICUAS)*, 2016.

[C15] B Shang, C Wu, T Meng, C Gao, Y Zhang. **A Data/Image Transmission Device Based on TCP/IP Protocol.** *8th International Conference on Wireless Communications, Networking and Mobile Computing (WiCOM)*, 2012.

[C16] T Meng, C Wu, B Shang, C Gao, Y Zhang. **Design of point to multi-point wireless communication system based on ZigBee.** *7th International Conference on Wireless Communications, Networking and Mobile Computing (WiCOM)*, 2011.

Annual Meeting Contributions

[M1] F Arnob, Bo Shang, Yiqiao Li, Abolfazl Karimpour. **Understanding Infrastructure-ADAS Interactions in Urban Crash Risk: A Multimodal AI Approach for Intersection Safety.** *TRB 105th Annual Meeting (Transportation Research Board)*, 2026. Record

[M2] Bo Shang, Yiqiao Li, Blanche Horbach, Kevin Villon. **Scalable Urban Truck Characterization using Deep Learning and Street View Imagery: A Drive-by Sensing Approach.** *TRB 105th Annual Meeting (Transportation Research Board)*, 2026. Record

[M3] Yiqiao Li, Jie Wei, Bo Shang, Camille Kamga. **Few-Shot Learning for Truck Classification from LiDAR Sensor using Vision-language Models.** *TRB 105th Annual Meeting (Transportation Research Board)*, 2026. Record

Preprints

[P1] Yiqiao Li, Bo Shang, Jie Wei. **Bridging the Modality Gap in Roadside LiDAR: A Training-Free Vision-Language Model Framework for Vehicle Classification.** *arXiv preprint arXiv:2602.09425*, 2026. arXiv DOI

[P2] Bo Shang, Yiqiao Li, Blanche Horbach, Kevin Villon. **Exploring Drive-By Sensing and Deep Learning for Urban Truck Characterization Using Google Street View Imagery.** *SSRN preprint 5312573*, 2025. DOI SSRN

Technical Reports

[R1] Yiqiao Li, Bo Shang, Weicong Feng, Jie Wei, Camille Kamga. **Investigation of Emerging Sensing and AI/ML Technologies to Enhance the Safety of Vulnerable Roadway Users at Signalized Intersection.** *U.S. Department of Transportation, University Transportation Centers Program, final technical report*, 2026. Published May 1, 2026; National Transportation Library record DOT 91974. PDF Record

[R2] Pengfei Ma, Tarutal Ghosh Mondal, Ying Zhuo, Zhenhua Shi, Bo Shang, Liujun Li, Genda Chen. **CAAP Final Report.** *Pipeline and Hazardous Materials Safety Administration (PHMSA), CAAP Final Report*, 2022. September 20, 2022; contract 693JK31950005CAAP; 109 pp. Record

[R3] Y Hu, J Yang, Y Cao, F Yang, B Shang. **Northeastern University Autonomous Aerial Robotics Team.** *International Aerial Robotics Competition (IARC) Symposium*, 2014.

[R4] Y Hu, J Yang, B Shang, S Liu. **Northeastern University Autonomous Aerial Robotics Team.** *International Aerial Robotics Competition (IARC) Symposium*, 2013.

Thesis

[T1] B Shang. **SLAM Research Based on ROS and Indoor Quadrotor.** *Northeastern University (M.S. Thesis)*, 2013.

Posters and Presentations

Posters

[PO1] B Shang, L Li, P Jiao, R Huerta, J Ressel, A Rawlings, B Scharfenberg, et al.. **Drone vision-based clamping strategy for bridge inspection.** *INSPIRE-UTC Poster*, 2021. Record

[PO2] B Shang, A Reven, P Jiao, B Li, G Chen. **Vision-Based Non-GPS UAV Guidance for Bridge Inspection.** *INSPIRE-UTC 2020 Annual Meeting (Poster)*, 2020.

[PO3] A Reven, P Jiao, B Shang, G Chen. **Clamping Design for Bridge Inspection Robot Deployment Systems (BIRDS) Prototype II.** *INSPIRE-UTC 2020 Annual Meeting (Poster)*, 2020.

Presentation Slides

[S1] A Reven, P Jiao, B Shang, G Chen. **Bridge Inspection Robot Deployment Systems (BIRDS) Prototype II.** *INSPIRE-UTC 2020 Annual Meeting (Slides)*, 2020.

Patents and Patent Applications

[PT1] Genda Chen, Alec Reven, Bo Shang, Zhenhua Shi, Liujun Li, Boyang Li. **Unmanned vehicle having flight configuration and surface traverse configuration.** *U.S. Patent 12,296,994 B2*, 2025. Granted May 13, 2025. Patent

[PT2] Chengdong Wu, Bo Shang, Yunzhou Zhang, Chengxi Gao, Tingting Meng. **Data/image transmission device based on TCP/IP (Transmission Control Protocol/Internet Protocol).** *Chinese Patent CN 102427464 B*, 2014. Granted April 30, 2014. Patent

[PT3] Yunzhou Zhang, Bo Shang, Chengdong Wu, Pengju Si. **Internet-based interactive digital media terminal device.** *Chinese Patent Application CN 102306237 A*, 2012. Patent application published January 4, 2012. Patent

Professional Service

Journal reviewer

Nonlinear Dynamics; Journal of Intelligent & Robotic Systems; ISA Transactions; IEEE Transactions on Control Systems Technology; Intelligent Buildings International; International Journal of Advanced Robotic Systems; Control Engineering Practice; IET Control Theory and Applications; Mechatronics.

Conference reviewer

International Conference on Unmanned Aircraft Systems (ICUAS); IEEE International Conference on Robotics and Automation (ICRA); IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI).

Session Chair | 2016

International Conference on Unmanned Aircraft Systems (ICUAS).

Review record

Web of Science profile

Honors, Support, and Credentials

Teaching Certificate, Association of College and University Educators (ACUE) | 2021

FAA Remote Pilot Certificate for Small Unmanned Aircraft Systems | 2016

China Scholarship Council (CSC) doctoral exchange support | 2015 - 2017

Two-year exchange program at the University of California, Merced; US$38,400 in financial support.

International Aerial Robotics Competition | 2014

Best System Control Award and Best Mission Planning Award, as team leader.

International Mathematical Contest in Modeling | 2010

Meritorious Prize, as programmer.

National Smart Car Competition, Freescale, China | 2010

First Prize, Northeastern Region.

Technical Expertise

Robotics and sensing

ROS / ROS 2, NVIDIA Jetson, Raspberry Pi, Arduino, embedded Linux; LiDAR, RGB/RGB-D, thermal, IMU, and acoustic sensing; sensor bring-up, synchronization, calibration, and field troubleshooting.

Software and machine learning

Python, C++, PyTorch, TensorFlow, OpenCV, Linux, Docker, Git/GitHub Actions, AWS; computer vision, 3D reconstruction, visual SLAM, model packaging, and data pipelines.

Evaluation and research delivery

Dataset and configuration versioning, labeling and data quality assurance, cross-site evaluation, replay and failure analysis, human review, GIS and traffic analytics, and technical documentation.